KEL159

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Teaching Note:
A&D High Tech (B): Managing Scope Change

Note: There are free supplementary materials not included in your purchase that are necessary to teach this case. Please e-mail cases@kellogg.northwestern.edu or call (847) 491-5400 for access to these materials.

Synopsis

The case is based upon a real $25 million project at a major U.S.-based computer manufacturer. For confidentiality reasons the company has been disguised as A&D High Tech. The Web-based online ordering system project is required by sales and marketing for the fall holiday season. If the project misses this window, the firm will lose substantial market share to competitors. A&D High Tech (B) takes place three months into the original project plan. The project manager has just been fired and the management challenge is to find out what is wrong with the project and recommend fixes. In addition, the scope of the project has changed: the VP of marketing has an additional promotional bundle requirement.

A&D High Tech (A) examines how to create and analyze a project plan in Microsoft Project. In order to make the case manageable for students we reduced the size of the project and the corresponding number of resources to approximately $1 million, but retained all features of the original project. Case B gives actual work done on each task three months into the project. Students must answer the management questions: Can the project be fixed and completed in time for the holiday season? Can the additional requirements be incorporated, and if so, what is the best approach?

In order to answer these questions, earned value data can be extracted from Microsoft Project and analyzed. These data provide important insights into the root cause of project problems. The next step is to reduce the scope of the project and reassign resources. However, one must be cognizant that indiscriminately adding people can slow a project down, not speed it up. Finally, the additional promotional bundle requirement from the marketing VP provides an important outsourcing management discussion.

These two cases, A&D High Tech (A) and (B), are used in the Kellogg MBA course entitled *Managing Technology Portfolios and Projects* and in multiple Kellogg executive programs. The case can also be taught using other project management software tools such as Primavera.

Purpose

The case teaches students how to analyze a project in trouble using Microsoft Project or other project management software tools. More important, the case teaches prospective executives how to manage a project in trouble by first accurately diagnosing what is wrong with the project, then reducing scope where necessary, and finally replanning the project with reallocated resources. In addition, students will learn the tradeoffs of outsourcing to highly specialized professionals versus average contractors.

The case centers on A&D High Tech's need to quickly develop an online store in response to a competitive disadvantage. Data for the entire plan are given to students, who should in turn analyze the plan using project management tools such as Gantt charts, critical path analysis, and earned value analysis. Students should be able to learn and practice using project planning and management software such as Microsoft Project.

In Case B, the CIO of A&D High Tech needs to determine whether the online store project can be completed in time for the holiday shopping season. The project manager needs to assess the current state of the project and also evaluate if the scope changes that were requested can be incorporated. The project manager has all the necessary details but needs to analyze them and determine the best course of action.

Case Questions

The case assignment is as follows:

Your team has been assigned to help Chris Johnson troubleshoot the online store project and develop a presentation and recommendation for Matt Webb, the CIO. Time is short. The current project actuals have been provided in Exhibit 1 of A&D High Tech (B), and are also available in the Excel file accompanying the case. A copy of the earned value analysis template is provided in Exhibit 2 of A&D High Tech (B) and also in the Excel file accompanying the case.

Additional related questions are:

1. *As of August 26, what is the projected cost and schedule? What are the trends? What is the critical path(s)?*

2. *Given the dimensions of project management and critical paths of a project, what can be done to bring the project back on track?*

3. *Is it possible to implement the changes to incorporate the promotional bundles and to complete the project by December 1? What are the necessary changes to the plan? Are there any additional risks involved in the revised plan?*

4. *What is the estimated cost variance for the project?*

An additional disclaimer is given to students, who may only focus on the answers to these questions: *These questions are intended to help your analysis. However, your final solution to the case and recommendations to management should not necessarily be limited to the answers to these questions or the assumptions in the case.*

Analysis

This analysis section looks at the project plan and determines the best course of action. We also quantitatively determine the project completion date and costs. In addition, the associated technical note "A&D High Tech: Microsoft Project Workshop" is available so that students can quickly learn the fundamentals of Microsoft Project. If project management software is not available, students can download Microsoft Project software for a free thirty-day trial from Microsoft.com to complete the case assignment.

FRAMEWORK FOR SOLVING THE CASE

Students will work through the case in groups. The solution to A&D High Tech (A) is typically given as a starting point for the students to work on A&D High Tech (B). They will likely approach the case by entering the information about the project into Microsoft Project. While it is possible to complete the case without the aid of project planning software, the exercise becomes much more tedious. Thus, learning to use this software is an integral part of the case. Using the software, students can easily tabulate metrics and costs, make "what-if" scenario changes, and export data into a spreadsheet to calculate the earned value information.

DETAILED SOLUTION

After entering the actual data into project planning software and the remaining estimated work for the tasks in the project plan, it is projected that the total cost of the project would be $898,419.89 and that the project would be completed and online by December 15, 2003. The single project-critical path is the task *Build ERP Interface*.

The next step is extracting earned value data from the project plan with actuals (see the last three slides of the accompanying slide presentation corresponding to the Microsoft Project Workshop technical note for instructions on how to do this).[1] See **Exhibit TN1** for a graph plotting key earned value parameters for the overall project that is easily extracted into Microsoft Excel from Microsoft Project: Actual Cost of Work Performed (ACWP), Budgeted Cost of Work Performed (BCWP), and the Budgeted Cost of Work Scheduled (BCWS). The earned value analysis using the BCWP, BCWS, and ACWP parameters as of August 26, 2003, shows that there is a large negative work variance and a small negative cost variance. In other words, the project is earning the money that is being spent, but is substantially behind plan. These data suggest a pattern that corresponds to the project not having enough staff.

The EVA template, Exhibit 2 in Case B, can be used to calculate the important earned value ratios for each component of the project: Cost Performance Index (CPI), Schedule Performance Index (SPI), and Control Ratio (CR). See **Exhibit TN2** for an excerpt of the earned value analysis from the solution file that accompanies this case. Cost or schedule performance indices less than 0.9 indicate a problem. The CR is the combination of the cost and schedule ratios, so a CR less than 0.9 or greater than 1.1 indicates a problem. A detailed review of the earned value analysis data for each of the subprojects in the program reveals that the initial problem occurred in the

[1] For an introduction to earned value project management, see Mark Jeffery, "Ariba Implementation at MED-X: Managing Earned Value," Case #5-404-763 (Kellogg School of Management, 2006). See also James P. Lewis, *Fundamentals of Project Management* (New York: AMACOM, 2001). This text, and especially Chapter 8, is required reading for the Kellogg MBA class.

requirements for the ERP interface. The earned value analysis data also highlights other issues in the project, such as *Build Shopping Cart*.

Note that analysis of the project software Gantt chart alone often does not give complete insight into the problems with a project. The earned value data analysis gives executives a detailed diagnostic tool to understand what is wrong with the project, and where things went wrong initially.

The following is the optimal solution in order to have the system completed by December 3, 2001. (Note: This equates to having the task *Deploy to Production* finish as of December 3, 2001, not the entire project having a finish date of that day.)

1. Reduce work for the task *Build Submit Order Pages & Components* by 50 percent. The task is now twelve person-days of work. This reduced scope was suggested in the case text.

2. Reassign Developer 2 from the task *Build Submit Order Pages & Components* to the task *Build ERP Interface*. Here we assume that the *Build Submit Order Pages & Components* has similar characteristics to *Build ERP Interface* so that Developer 2 can quickly get up to speed, avoiding the "mythical man-month."[2]

3. Add new tasks for the promotional bundles, assuming that the company uses Microsoft consultants for the configuration work and the current team for testing.

Task ID	Task Name	Work Estimate (days)	Duration (days)	Constraint	Predecessors	Resource Name
61	Promotional Bundle					
62	Promotional Bundle Configuration[a]	3	3	start no earlier than 9/11/03 (two weeks from 8/26/03)		Microsoft Consultant
63	Promotional Bundle Testing[b]	24	3		62, 55	Kara Siposki (Test Lead), Todd Eliason (Tester), Marc Sanders (Development Lead), Developer 1 (TBD), Developer 2 (TBD), Developer 3 (TBD), Ryan Neff (Functional Lead), Stacy Lyle (Functional Analyst)

[a] Start: Tue. 9/9/03, Finish: Thurs. 9/11/03

[b] Start: Tue. 11/11/03, Finish: Fri. 11/14/03

Note: Notice that Task 63 is dependent on Task 55, *Validation Testing*.

[2] Frederick P. Brooks, *The Mythical Man-Month: Essays on Software Engineering* (Reading, MA: Addison-Wesley, 1995).

4. Assigned a $500 per hour rate for the Microsoft consultant. As we will see in the following sections the Microsoft consultant, with a high hourly wage, is actually cheaper and less risky than the alternative.

5. Add Task 61 as a dependency for the *Deployment* roll-up task.

6. Reduce the number of remaining work days for *Manage Project* to 55.

Summary. If these steps are taken the online store will be implemented by December 1, 2003. The projected cost variance is approximately $65,199.89, or approximately 7.7 percent above baseline cost.

Case Question Solutions

The assignment can be completed by answering the following questions:

1. As of August 26, what is the projected cost and schedule? What are the trends? What is the critical path(s)?

As of the case date, the projected cost is $898,419.89 and the system would be implemented by December 15, 2003. The earned value analysis trends suggest that there is a large negative work variance and a small negative cost variance. Therefore, the project is lagging in the work performed at roughly the expected cost for this work. It becomes clear if you extract earned value analysis from Microsoft Project into Microsoft Excel, the critical path lies in the design and development of the ERP interface (details on how to extract earned value analysis data from Microsoft Project can be found in “A&D High Tech: Microsoft Project Workshop”).

2. Given the dimensions of project management and critical paths of a project, what can be done to bring the project back on track?

According to the golden triangle of project management, there are three levers that can influence projects: time, resources, and scope. Since the goal is to keep the project on track (time), solutions could include increasing resources or reducing scope (or doing both) in order to reduce the time the project takes.

3. Is it possible to implement the changes to incorporate the promotional bundles and to complete the project by December 1? What are the necessary changes to the plan? Are there any additional risks involved in the revised plan?

Yes, it is possible to implement the changes for the promotional bundles and complete the project by December 1. This would require hiring additional expert resources and cutting certain project scope features that are no longer needed. This is probably the least risky approach with regard to resources, since it basically keeps the current staff, other than the Microsoft consultant. However, the new modified plan has absolutely no slack to make the date. Thus, any additional delays would cause the project to miss the December 1 deadline.

4. What is the estimated cost variance for the project?

The estimated cost variance for the project, given the above changes, will be $65,199.89, or approximately 7.7 percent above baseline cost.

Teaching the Case

COMMON STUDENT APPROACHES TO THE SOLUTION

All student groups should be able to enter the data and calculate the total costs and project completion date. Using project planning software, it should be fairly straightforward for students to study the Gantt chart and highlight the critical paths. Students should also be able to perform an earned value analysis from the data extracted from Microsoft Project.

It is likely the students will make three mistakes:

1. Entering in the data incorrectly. This can happen in a variety of places. The technical note "A&D High Tech: Microsoft Project Workshop" is intended to help students who are unfamiliar with Microsoft Project. It is recommended that students receive a basic tutorial on Microsoft Project prior to assigning the case.

2. Taking the project finish date as the project completion date. The project includes a "project wrap-up" that takes an additional ten days after the system has been deployed. The ten-day wrap-up period can be subtracted from the total project to give executives the date the system will go into production.

3. Not extracting earned value data from Microsoft Project to Microsoft Excel and completing a thorough diagnosis of the project. The technical note "A&D High Tech: Microsoft Project Workshop" has explicit instructions on how to successfully extract the data.

TEACHING APPROACH

Students discuss the case in groups and may be asked to prepare a few slides to present to Chris Johnson, the new interim project manager. In the MBA class at Kellogg, one team is usually selected to present the case. Students are encouraged to question the team presenting as if they were Johnson and the senior executive team. This can create an interesting dialogue, with the students enjoying their roles. Following the group presentation, the instructor facilitates a discussion and presents an optimal solution. In the Kellogg executive program, time is more limited, so the debrief consists of a thirty-minute facilitated discussion.

At this point, the instructor can walk the class through the solution to the case. Any debrief should start by analyzing the project plan as of August 26, the date when the "actuals" were given. Use the Gantt chart to show that the critical path is the ERP interface. However, the Microsoft Project Gantt chart view is deceptive; it does not clearly help diagnose what went wrong with the project. The instructor can then show a graph with the plot of ACWP, BCWP, and BCWS on a time trend (see Exhibit TN1). The earned value analysis depicts a large negative work variance and a small negative cost variance. This pattern indicates that although work is being performed according to the planned cost, it is not being performed according to the planned schedule. In principle, this simply suggests adding more people, but students should be aware of the danger in offering this as a solution: adding more people indiscriminately to a project in trouble typically slows the project down even more instead of speeding it up (the mythical man-month).[3]

The instructor can then highlight the point that building the baseline plan is an important exercise, since this analysis would not have been possible without it. Moreover, the act of designing and thinking about the project gives the project manager insight into dependencies and risks. Clearly the project evolved in a very different manner than the original baseline plan.

In order to get a better idea what portion of the project is actually the cause of the delay, extracting the earned value data from Microsoft Project to Microsoft Excel is a good idea. Note that by extracting the data, the students can quickly scan the schedule impact, cost impact, and control ratios to determine the specific issues. See Exhibit TN2 for an excerpt of the earned value analysis from the file that accompanies this case. A quick scan of the exported earned value data reveals that the *Updating and Calculating Shopping Cart* task has an SPI of 1.0 and CPI and CR of 0.75. Furthermore, the *Create ERP Interface Requirements* task has an SPI of 0.67, a CPI of 0.47, and CR of 0.31. The *Design ERP Interface* task has an SPI of 0.67, a CPI of 0.67, and CR of 0.44. Finally, the *Build ERP Interface* task has an SPI of 0.07, a CPI of 0.56, and CR of 0.04. Although there is a small problem with the shopping cart task, the majority of the problems are with the ERP-related tasks. The earned value data shows how the real problem is with the ERP requirements, and this problem cascaded through the project plan. The overall project has an SPI of 0.67, a CPI of 0.94, and CR of 0.63.

Next we can ask, "When will the project be done?" First we calculate the estimate at complete (EAC). This will be a ballpark estimate. We know that the overall project SPI is 0.67 and that the total estimated work days is 127. The EAC is simply calculated from the ratio 127/0.67. This gives us an expected project length of 189 days, or approximately 62 additional work days. Thus, the completion of the project is slated for sometime in March, far beyond the holiday season launch goal. Using the overall project SPI we can see the project in its current form will not be completed in time. Now that we have identified the specific issues with the project, the instructor can move on to address these specific problems.

In thinking through solutions for the various project issues, stress to students that managing the critical path and related dependencies is the key to success. To this point, cutting the scope of the project is appropriate, given the circumstances. Target areas for cutting scope are the shopping cart functionality and the *Build Submit Order Pages & Components* task. By reducing the work for the task *Build Submit Order Pages & Components* by 50 percent, the task is now twelve person-days of work. The reduced amount of work associated with the shopping cart functionality will allow the project manager to shuffle a developer from that task to the more critical ERP interface task. Here we assume that the *Build Submit Order Pages & Components* has similar characteristics to *Build ERP Interface* so that Developer 2 can quickly get up to speed, avoiding the mythical man-month. This new work assignment "pulls" the critical path back, so the ERP interface is no longer the long pole.

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[3] Ibid.

Next, the project manager must deal with the increase in scope of the project. Outsourcing the promotional bundle configuration work seems obvious. Since the technical lead suggested that his development team is getting too big, it is not wise to simply add contract developers. This is consistent with avoiding the mythical man-month problem. In addition, it is unclear whether a new contractor would have the right skills to do this key piece of work correctly. These two uncertainties add to the risk of further delays. Given the fact that it only takes two weeks to arrange for the Microsoft consultant to arrive, development costs can be kept low. The decision also allows the existing development schedule to remain intact by having this work done in parallel.

There is an interesting discussion concerning outsourcing to the Microsoft consultant versus the more general consultants. The Microsoft consultant's rate is $500 per hour, much more than the general consultants'. Students are often reluctant to use the Microsoft consultant because the hourly rate is so high. However, the total cost of the Microsoft consultant is actually less than that of the general contract consultant. This is because the Microsoft consultant will accomplish the task in only two days. This highlights the difference in performance between different developers. See **Exhibit TN3** for software developer performance curves. The takeaway for students is that when a project is in trouble, bringing in highly experienced developers is a good strategy since their productivity can be as much as fourteen times greater than that of an average developer. The high cost of these experts can be offset by their extreme productivity.

For many of the same reasons, the testing portion was to be performed by the original testing team, not a team composed of new resources. The testing of the promotional bundle was to happen after the already-planned testing phases. This further mitigates risk in trying to incorporate and test the extra functionality into the previously planned tasks.

At this point in the debrief, the instructor can facilitate a discussion on what the students see as the risks going forward. The point should be made that since there is absolutely no slack in the schedule, any delay in a task would need to be made up by a subsequent task. Moreover, as with the original project plan, there is no contingency in the plan.

Next, the instructor can give a quick lecture on suggestions for what to do if the students find themselves managing a project faced with numerous difficulties. One key point is the variance in skill level of IT workers. When feasible and useful, project managers should consider bringing in experts, because they can be as much as fourteen times more effective than an average person (see Exhibit TN3).

Students should also remain cautious when using overtime to "catch up" projects that are behind schedule. As a general rule, managers have a window of about two weeks of overtime to get a project back on schedule. Attempting to sustain overtime longer than that will result in a significant drop in productivity, as the employees will culturally adapt to the new schedule. Employees may show up early and leave late, but more time will be wasted during the day (e.g., two-hour lunch breaks or two hours offsite working out) and morale will ultimately suffer, resulting in a worse scenario for the project manager.

Another tool for project managers is the use of bonuses. These bonuses should be used to entice the best workers to put forth their best effort on a particular project. Bonuses should be paid based on performance and not on the successful completion of the project. Employees have no control over the efforts of others, and not paying the bonus will result in future motivation problems. Finally, project managers can place some idle workers on other billable projects. This will reduce the costs of the project, which can be used at a later time if needed.

Finally, the instructor can give the students one last takeaway. The following Frederick P. Brooks quote in *The Mythical Man-Month* is a good analogy for most projects: “When one hears of disastrous schedule slip in project, [one] imagines that a series of major calamities must have befallen it. Usually, however, the disaster is due to termites, not tornadoes . . . the schedule slips one day at a time.” For a comprehensive set of slides for facilitating the case, see the PowerPoint file accompanying this teaching note.

Executive Expert Comments

As with “A&D High Tech (A): Managing Projects for Success,” Case B is based on a real project with a budgeted cost of approximately $25 million and staffed with 150 people at its peak. Unlike the case, however, the problems that arose were not because of the lack of project management metrics or techniques being employed. As mentioned in the teaching note for A&D High Tech (A), the main problem for the system turned out to be the performance, but this was not the full story.

One of the senior project managers from the consulting firm was severely disguising the numbers to get the project to appear on track. He was in line to make partner in the firm and did not want the project to “look bad.” Once this was revealed, this manager and many of his lieutenants were fired from the project and subsequently from the consulting firm. A complete audit of the project metrics found that much of the work in the plan had not been performed. This manager was up for promotion and wanted to show that he could deliver a project on time and under cost. Luckily for the consulting company, as in the case, cost overruns were minor. The new managers were able to put more resources onto the team to get the project back on track. However, the project was significantly delayed and did not make the holiday shopping season, which was a strategic imperative for the client.

Even though the problem was related to a single person, the consulting firm ruined its reputation with the client as a dependable partner. The client had done over $200 million in business with the consulting firm over the prior seven years but chose to effectively terminate the entire relationship. The lesson here is that even sophisticated project management and analysis techniques cannot replace a manager who simply chose not to tell the truth.

Exhibit TN1: Earned Value Analysis

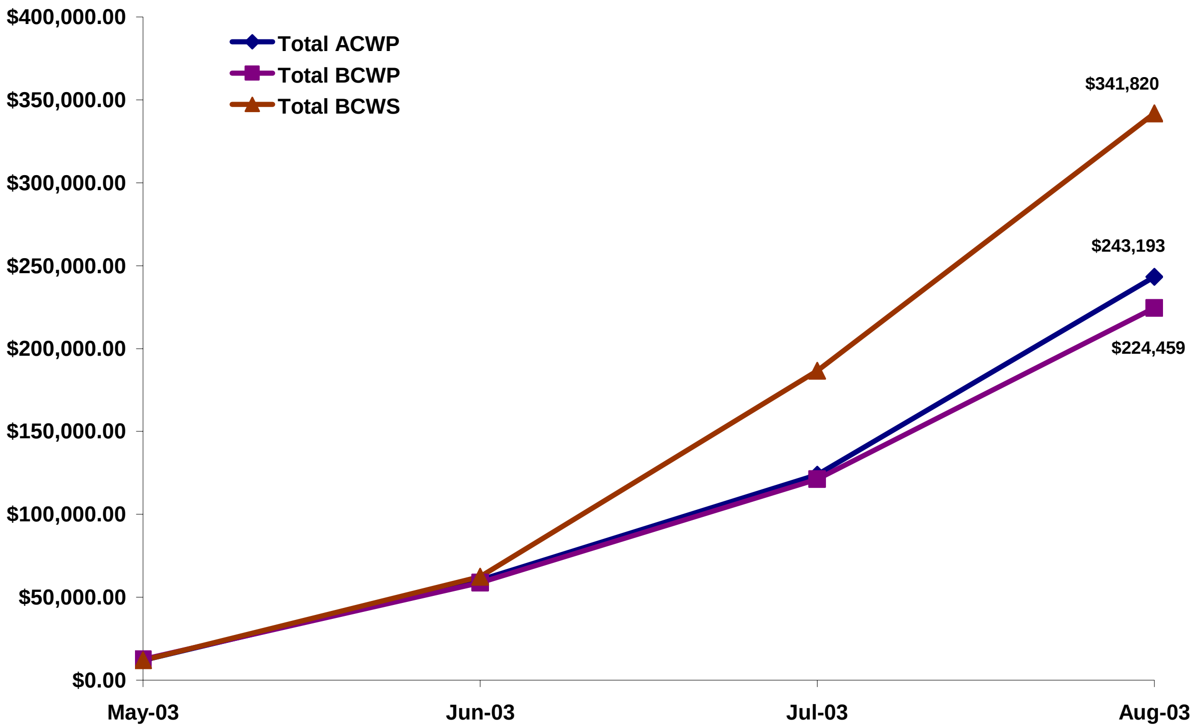

Exhibit TN2: Earned Value Analysis Excerpt

Months	***5/1/2003***	***6/1/2003***	***7/1/2003***	***8/1/2003***
Overall Project				
ACWP	$0.00	$6,600.00	$12,000.00	$31,200.00
BCWP	$0.00	$6,856.00	$9,376.00	$22,176.00
BCWS	$0.00	$7,800.00	$22,320.00	$30,000.00
SPI		0.88	0.42	0.74
CPI		1.04	0.78	0.71
CR		0.91	0.33	0.53
Project Management				
System Requirements				
Identify Technical Infrastructure Needs				
ACWP	$0.00	$600.00	$600.00	$600.00
BCWP	$0.00	$1,200.00	$1,200.00	$1,200.00
BCWS	$0.00	$1,200.00	$1,200.00	$1,200.00
SPI		1.00	1.00	1.00
CPI		2.00	2.00	2.00
CR		2.00	2.00	2.00
Software Requirements				
Updating and Calculating Shopping Cart				
ACWP	$0.00	$2,400.00	$2,400.00	$2,400.00
BCWP	$0.00	$1,800.00	$1,800.00	$1,800.00
BCWS	$0.00	$1,800.00	$1,800.00	$1,800.00
SPI		1.00	1.00	1.00
CPI		0.75	0.75	0.75
CR		0.75	0.75	0.75
Taking Payments				
ACWP	$0.00	$3,000.00	$3,000.00	$3,000.00
BCWP	$0.00	$3,576.00	$3,576.00	$3,576.00
BCWS	$0.00	$3,600.00	$3,600.00	$3,600.00
SPI		0.99	0.99	0.99
CPI		1.19	1.19	1.19
CR		1.18	1.18	1.18
Create ERP Interface Requirements				
ACWP	$0.00	$600.00	$6,000.00	$6,000.00
BCWP	$0.00	$280.00	$2,800.00	$2,800.00
BCWS	$0.00	$1,200.00	$4,200.00	$4,200.00
SPI		0.23	0.67	0.67
CPI		0.47	0.47	0.47
CR		0.11	0.31	0.31
Detailed Design				
Design ERP Interface				
ACWP	$0.00	$0.00	$0.00	$19,200.00
BCWP	$0.00	$0.00	$0.00	$12,800.00
BCWS	$0.00	$0.00	$11,520.00	$19,200.00
SPI				0.67
CPI				0.67
CR				0.44

Note: Highlighted cells indicate major positive and negative impacts to the project.

Exhibit TN3: Performance Curves